Natural killer cells at the forefront of cancer immunotherapy with immune potency, genetic engineering, and nanotechnology.

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Natural killer (NK) cells are vital components of the human immune system, acting as innate lymphocytes and playing a crucial role in immune surveillance. Their unique ability to independently eliminate target cells without antigen contact or antibodies has sparked interest in immunological research. This review examines recent NK cell developments and applications, encompassing immune functions, interactions with target cells, genetic engineering techniques, pharmaceutical interventions, and implications in cancers. Insights into NK cell regulation emerge, with a focus on promising genetic engineering like CAR-engineered NK cells, enhancing specificity against tumors. Immune checkpoint inhibitors also enhance NK cells' potential in cancer therapy. Nanotechnology's emergence as a tool for targeted drug delivery to improve NK cell therapies is explored. In conclusion, NK cells are pivotal in immunity, holding exciting potential in cancer immunotherapy. Ongoing research promises novel therapeutic strategies, advancing immunotherapy and medical interventions.

Application of natural killer cells in pancreatic cancer.

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Pancreatic cancer, a highly malignant disease, is characterized by rapid progression and early metastasis. Although the integrative treatment of pancreatic cancer has made great progress, the prognosis of patients with advanced pancreatic cancer remains extremely poor. In recent years, with the advancements in tumor immunology, immunotherapy has become a promising remedy for pancreatic cancer. Natural killer (NK) cells are the key lymphocytes in the innate immune system. NK cell function does not require antigen pre-sensitization and is not major histocompatibility complex restricted. By targeting tumors or virus-infected cells, the cells play a key role in immune surveillance. Although several questions about NK cells in pancreatic cancer still need to be further studied, there are extensive theories supporting the clinical application prospects of NK cell immunotherapy in pancreatic cancer. Since very few studies have evaluated the role of NK cells in pancreatic cancer, this review provides a comprehensive update of the role of NK cells in pancreatic cancer immunotherapy.

NK cell-based cancer immunotherapy: from basic biology to clinical development.

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Natural killer (NK) cell is a specialized immune effector cell type that plays a critical role in immune activation against abnormal cells. Different from events required for T cell activation, NK cell activation is governed by the interaction of NK receptors with target cells, independent of antigen processing and presentation. Due to relatively unsophisticated cues for activation, NK cell has gained significant attention in the field of cancer immunotherapy. Many efforts are emerging for developing and engineering NK cell-based cancer immunotherapy. In this review, we provide our current understandings of NK cell biology, ongoing pre-clinical and clinical development of NK cell-based therapies and discuss the progress, challenges, and future perspectives.

Modulation of intrinsic inhibitory checkpoints using nano-carriers to unleash NK cell activity.

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Natural killer (NK) cells provide a powerful weapon mediating immune defense against viral infections, tumor growth, and metastatic spread. NK cells demonstrate great potential for cancer immunotherapy; they can rapidly and directly kill cancer cells in the absence of MHC-dependent antigen presentation and can initiate a robust immune response in the tumor microenvironment (TME). Nevertheless, current NK cell-based immunotherapies have several drawbacks, such as the requirement for ex vivo expansion of modified NK cells, and low transduction efficiency. Furthermore, to date, no clinical trial has demonstrated a significant benefit for NK-based therapies in patients with advanced solid tumors, mainly due to the suppressive TME. To overcome current obstacles in NK cell-based immunotherapies, we describe here a non-viral lipid nanoparticle-based delivery system that encapsulates small interfering RNAs (siRNAs) to gene silence the key intrinsic inhibitory NK cell molecules, SHP-1, Cbl-b, and c-Cbl. The nanoparticles (NPs) target NK cells in vivo, silence inhibitory checkpoint signaling molecules, and unleash NK cell activity to eliminate tumors. Thus, the novel NP-based system developed here may serve as a powerful tool for future NK cell-based therapeutic approaches.

INTASYL self-delivering RNAi decreases TIGIT expression, enhancing NK cell cytotoxicity: a potential application to increase the efficacy of NK adoptive cell therapy against cancer.

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Natural killer (NK) cells are frontline defenders against cancer and are capable of recognizing and eliminating tumor cells without prior sensitization or antigen presentation. Due to their unique HLA mismatch tolerance, they are ideal for adoptive cell therapy (ACT) because of their ability to minimize graft-versus-host-disease risk. The therapeutic efficacy of NK cells is limited in part by inhibitory immune checkpoint receptors, which are upregulated upon interaction with cancer cells and the tumor microenvironment. Overexpression of inhibitory receptors reduces NK cell-mediated cytotoxicity by impairing the ability of NK cells to secrete effector cytokines and cytotoxic granules. T-cell immunoreceptor with immunoglobulin and ITIM domains (TIGIT), a well-known checkpoint receptor involved in T-cell exhaustion, has recently been implicated in the exhaustion of NK cells. Overcoming TIGIT-mediated inhibition of NK cells may allow for a more potent antitumor response following ACT. Here, we describe a novel approach to TIGIT inhibition using self-delivering RNAi compounds (INTASYL™) that incorporates the features of RNAi and antisense technology. INTASYL compounds demonstrate potent activity and stability, are rapidly and efficiently taken up by cells, and can be easily incorporated into cell product manufacturing. INTASYL PH-804, which targets TIGIT, suppresses TIGIT mRNA and protein expression in NK cells, resulting in increased cytotoxic capacity and enhanced tumor cell killing in vitro. Delivering PH-804 to NK cells before ACT has emerged as a promising strategy to counter TIGIT inhibition, thereby improving the antitumor response. This approach offers the potential for more potent off-the-shelf products for adoptive cell therapy, particularly for hematological malignancies.

Natural killer cell engineering - a new hope for cancer immunotherapy.

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Natural killer (NK) cells are an important component of the innate immune system, particularly for metastasis immunosurveillance. They can rapidly recognize and kill transformed cells without the requirement of specific neo-antigen recognition. Their effector functions are modulated by a range of stimulatory and inhibitory surface receptors that regulate their cellular activation, differentiation and homeostasis. However, cancer cells can evade NK cell detection by receptor interaction or secretion of soluble immunosuppressant molecules. Therefore, genetic reprogramming of these immune suppressing or activating receptors of NK cells is a promising strategy to augment NK cell tumoricidal functions. In this review, we highlight the current clinical trials of chimeric antigen receptor engineered NK cells with redirected antigen specificity to eliminate hematological cancers and solid tumors. New alternative strategies that are advancing NK cell engineering for cancer treatment are also outlined. Lastly, different NK cell transgenesis approaches are reviewed and compared, and we discuss how these methods can be employed to maximize their anti-tumor effector functions.

Prognostic impact of natural killer cell recovery on minimal residual disease after autologous stem cell transplantation in multiple myeloma.

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Natural killer cells are a potent effector lymphocyte subset that can induce cytotoxicity without the need for antigen sensitization or presentation. NK cells are a tempting target -for immune therapy, monoclonal antibody, or genetic engineering-to enhance immune surveillance mechanisms against myeloma cells. We hypothesized an association between natural killer cell recovery after autologous stem cell transplantation (ASCT) and disease outcomes in multiple myeloma patients. We concluded a prospective study that started enrolling patients in January 2020 to identify the association between absolute NK cell count two to three after ASCT and disease outcomes after autologous stem cell transplantation in multiple myeloma using univariate and multivariate analysis. Natural killer cell recovery was evaluated during the third month after ASCT, day +60 to +90 post-ASCT. Our patients had a mean NK cell count of 90.53, ranging from 14 to 282 Cell/µL (Std Dev 84.64 Cell/µL). The odds of having a minimal residual disease (MRD-positivity) among patients with partial remission before transplantation is four times higher than patients with very good partial response or better (95% confidence interval 0.45-35.79). Our patients were classified into two groups based on MRD status after ASCT, an MRD-negative group of eight participants and an MRD-positive group of seven participants. The mean absolute NK cell count was significantly higher in the MRD-negative cohort, 131.38 Cell/µL, versus 43.86 Cell/µL in the MRD-positive group ($p = 0.049$). We conclude that for multiple myeloma patients treated with ASCT, high absolute NK cell counts two to three months after ASCT is an independent predictor for MRD negativity.

Application of natural killer immunotherapy in blood cancers and solid tumors.

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Natural killer (NK) cells are innate lymphoid cells characterized by their ability to attack aberrant and cancerous cells. In contrast to the activation of T-cells, NK cell activation is controlled by the interaction of NK cell receptors and their target cells in a manner independent of antigen organization. Due to NK cells' broad array of activation cues, they have gained great attention as a potential therapeutic agent in cancer immunotherapy. Ex vivo activation, expansion, and genetic modifications, such as the addition of a chimeric antigen receptor (CAR), will allow the next generation of NK cells to enhance cytotoxicity, promote survival, and create "off-the-shelf" products. In addition to these that are poised to greatly enhance their clinical activity, the inherent lack of potential for causing graft-versus-host disease (GVHD) and cytokine release syndrome (CRS) suggest that CAR NK cells have the potential to be complementary to CAR-T cells as a component of therapeutic strategies for cancer. In this review, we will provide a general understanding of NK cell biology, CAR-NK cell advantages over CAR-T cell therapy, barriers to making NK cell immunotherapy viable, and current NK cell clinical trials for hematological malignancies and solid tumors. The next generation of NK cells has potential to change the circumstances guiding present cancer immunotherapies.

Harnessing Natural Killer Immunity in Metastatic SCLC.

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SCLC is the most aggressive subtype of lung cancer, and though most patients initially respond to platinum-based chemotherapy, resistance develops rapidly. Immunotherapy holds promise in the treatment of lung cancer; however, patients with SCLC exhibit poor overall responses highlighting the necessity for alternative approaches. Natural killer (NK) cells are an alternative to T cell-based immunotherapies that do not require sensitization to antigens presented on the surface of tumor cells. We investigated the immunophenotype of human SCLC tumors by both flow cytometry on fresh samples and bioinformatic analysis. Cell lines generated from murine SCLC were transplanted into mice lacking key cytotoxic immune cells. Subcutaneous tumor growth, metastatic dissemination, and activation of CD8+ T and NK cells were evaluated by histology and flow cytometry. Transcriptomic analysis of human SCLC tumors revealed heterogeneous immune checkpoint and cytotoxic signature profiles. Using sophisticated, genetically engineered mouse models, we reported that the absence of NK cells, but not CD8+ T cells, substantially enhanced metastatic dissemination of SCLC tumor cells in vivo. Moreover, hyperactivation of NK cell activity through augmentation of interleukin-15 or transforming growth factor-β signaling pathways ameliorated SCLC metastases, an effect that was enhanced when combined with antiprogrammed cell death-1 therapy. These proof-of-principle findings provide a rationale for exploiting the antitumor functions of NK cells in the treatment of patients with SCLC. Moreover, the distinct immune profiles of SCLC subtypes reveal an unappreciated level of heterogeneity that warrants further investigation in the stratification of patients for immunotherapy.

Phase I study of safety and efficacy of allogeneic natural killer cell therapy in relapsed/refractory neuroblastomas post autologous hematopoietic stem cell transplantation.

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Despite low incidence, neuroblastoma, an immunologically cold tumor, is the most common extracranial solid neoplasm in pediatrics. In relapsed/refractory cases, the benefits of autologous hematopoietic stem cell transplantation (auto-HSCT) and other therapies are limited. Natural killer (NK) cells apply cytotoxicity against tumor cells independently of antigen-presenting cells and the adaptive immune system. The primary endpoint of this trial was to assess the safety of the injection of allogenic, ex vivo-expanded and primed NK cells in relapsed/refractory neuroblastoma patients after auto-HSCT. The secondary endpoint included the efficacy of this intervention in controlling tumors. NK cells were isolated and primed ex vivo (by adding interleukin [IL]-2, IL-15, and IL-21) in a GMP-compliant CliniMACS system and administered to four patients with relapsed/refractory MYCN-positive neuroblastoma. NK cell injections (1 and 5 × 107 cells/kg in the first and second injections, respectively) were safe, and no acute or sub-acute adverse events were observed. During the follow-up period, one complete response (CR) and one partial response (PR) were observed, while two cases exhibited progressive disease (PD). In follow-up evaluations, two died due to disease progression, including the case with a PR. The patient with CR had regular growth at the 31-month follow-up, and another patient with PD is still alive and receiving chemotherapies 20 months after therapy. This therapy is an appealing and feasible approach for managing refractory neuroblastomas post-HSCT. Further studies are needed to explore its efficacy with higher doses and more frequent administrations for high-risk neuroblastomas and other immunologically cold tumors.Trial registration number: irct.behdasht.gov.ir (Iranian Registry of Clinical Trials, No. IRCT20201202049568N1).